

Wandrille Ruffenach

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Research Interests

- **Dispersive PDEs:** ill-posedness and probabilistic well-posedness for nonlinear dispersive and wave equations. Spontaneous stochasticity in fractional nonlinear Schrodinger equation.
- **Physically relevant dynamical consequences of roughness:** spontaneous stochasticity, anomalous dissipation, intermittency.
- **Modeling of turbulence:** stochastic models of turbulence including intermittency and temporal statistics. Toy model of turbulent flows.

Current Appointment and Education

Ph.D. candidate in Physics

Oct. 2024–present

ENS de Lyon and Laboratoire de Physique de l'ENS de Lyon

Expected completion: July 2027. Advisors: Laurent Chevillard and Nikolay Tzvetkov.

M.Sc. & B.Sc. in Physics

Sept. 2020–July 2023

ENS de Lyon

Both degrees awarded with highest honors. Multidisciplinary training in theoretical physics, applied mathematics, and scientific computing.

Peer-Reviewed Articles

W. Ruffenach and L. Chevillard. “The spatio-temporal statistical structure of the turbulent dissipation field and its stochastic representation as a Gaussian multiplicative chaos.” *Comptes Rendus. Physique* **27**, 275–305 (2026)

M. Chatelain, J. Domingues Lemos, **W. Ruffenach**, M. Bourgoïn, C.-E. Bréhier, L. Chevillard, I. Sibgatullin, and R. Volk. “A spatio-temporal random synthetic turbulent velocity field: The underlying Gaussian structure.” *Journal of Fluid Mechanics* **1030**, A23 (2026)

W. Ruffenach, L. Fery, and B. Dubrulle. “Toy model of turbulent shear flow using vortons.” *Physical Review Fluids* **10**, 054601 (2025)

G. Beck, C.-E. Bréhier, L. Chevillard, R. Grande, and **W. Ruffenach**. “Numerical simulations of a stochastic dynamics leading to cascades and loss of regularity: Applications to fluid turbulence and generation of fractional Gaussian fields.” *Physical Review Research* **6**, 033048 (2024)

W. Ruffenach, L. Galantucci, and C. F. Barenghi. “Superfluid drain vortex.” *Journal of Low Temperature Physics* **212**, 375–388 (2023)

Proceedings

G. Beck, C.-E. Bréhier, L. Chevillard, R. Grande, and **W. Ruffenach**. “A linear model of dynamical sea wavenumber spectra.” in *Stochastic Transport in Upper Ocean Dynamics IV*, Mathematics of Planet Earth **15**, 15–50, Springer (2026), doi:10.1007/978-3-032-12749-5_2

Preprints

W. Ruffenach. “Spontaneous stochasticity and anomalous dissipation in collapsing wave turbulence.” arXiv:2607.18788 (2026)

W. Ruffenach, E. Simonnet, and N. Valade. “A measure-theoretic approach to spontaneous stochasticity.” arXiv:2607.16328 (2026)

W. Ruffenach and N. Tzvetkov. “Numerical study of probabilistic well-posedness of one-dimensional fractional nonlinear wave equations.” arXiv:2604.05938 (2026)

W. Ruffenach, E. Simonnet, and N. Valade. “Spontaneous stochasticity in the Armstrong–Vicol passive scalar.” [arXiv:2509.15683](https://arxiv.org/abs/2509.15683) (2025)

Oral Presentations

The spatio-temporal statistical structure of turbulent dissipation and its stochastic modeling June 2026
Opening on Small Scales (OSS 2026), Institut Camille Jordan, Lyon, France.
The spatio-temporal statistical structure of turbulent dissipation and its stochastic modeling May 2026
Turbulence Day, ENS de Lyon, Lyon, France.
A linear and stochastic model of energy cascade June 2025
Annual Meeting of the GDR Turbulence, Marseille, France.
A linear and stochastic model of energy cascade May 2025
Theory Day, Lyon, France.

Poster Presentations

Synthetic turbulence beyond Gaussianity: A causal model for space-time intermittency June 2026
Annual Meeting of the GDR Turbulence, Toulouse, France.
Spontaneous stochasticity in the Armstrong–Vicol passive scalar June 2026
Mathematics of Multiscale Modeling, IMPA, Rio de Janeiro, Brazil.
Spontaneous stochasticity in the Armstrong–Vicol passive scalar March 2026
Rencontres du Non-Linéaire, Paris, France.
Spontaneous stochasticity and anomalous dissipation in collapsing wave turbulence October 2025
Bridges over Turbulent Matters, Fall School, IESC, Cargèse, France.
A linear and stochastic model of energy cascade April 2025
Mathematical Fluid Dynamics, Summer School, IESC, Cargèse, France.
Cheap turbulence modeling with vortex particles March 2024
Rencontres du Non-Linéaire, Paris, France.

Teaching and Mentoring

Introduction to signal and image processing 2024–2026
ENS de Lyon
Teaching for undergraduate students.
Preparation for the French national *agrégation* examination 2024–2026
ENS de Lyon
Training and academic support for undergraduate students preparing the national competitive examination for secondary-school teaching.
Mentoring in the CPES undergraduate program 2024–2026
ENS de Lyon
Individual and small-group mentoring in the multidisciplinary *Cycle pluridisciplinaire d’études supérieures*.
Tutoring in quantum mechanics and mathematics 2024–2026
ENS de Lyon
Academic support for undergraduate students.
Oral examiner in physics 2021–2023
Les Chartreux, Lyon
Weekly oral examinations for first- and second-year students in the French *classes préparatoires* (PCSI, PC, and PSI).